

Extra Dimensions De-Literalised: Internal Modular Geometry on $X(2)$ as the 4D Replacement for Compactified Spacetime Coordinates

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Related deposits:

LoNalogy framework (v87.1): [doi:10.5281/zenodo.19115338](https://doi.org/10.5281/zenodo.19115338)

Yang–Mills proof (v90): [doi:10.5281/zenodo.19183838](https://doi.org/10.5281/zenodo.19183838)

Companion papers: Statistical mechanics on $X(2)$, Particle cosmology (v9), Hubble
reconstruction (v2), Coronal heating

Abstract

String theory requires extra spatial dimensions for quantum-gravitational consistency. Despite extensive searches at the LHC ($M_D \gtrsim 5.5\text{--}11.2\text{ TeV}$ for ADD graviton emission, $m_1 \gtrsim 2.2\text{--}5.6\text{ TeV}$ for RS Kaluza–Klein gravitons) and in short-distance gravity experiments ($R < 44\text{ }\mu\text{m}$ for $\delta = 1$), no evidence for extra dimensions has been found.

We show that in the LoNalogy framework, literal extra spacetime coordinates are unnecessary. The core action is 4-dimensional; the roles played by extra dimensions in string theory—gauge unification, family structure, coupling hierarchy—are assumed by the internal modular/fiber geometry of $X(2)$: the rank-5 bundle, three-cusp family, and bridge/hidden sectors. The integers $\nu_{\text{br}} = 12$, $d_{\text{fib}} = 7$, $N = 5$ that appear in LoNalogy are internal bundle dimensions, not spacetime coordinates.

The conclusion is that extra dimensions are not denied but *de-literalised*: they are reinterpreted as internal geometry of a 4D law, consistent with the mainstream AdS/CFT reading of warped extra dimensions as 4D composite dynamics.

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1 Introduction

Since Kaluza (1921) and Klein (1926), the idea that spacetime may have more than four dimensions has been a recurring theme in theoretical physics. String theory, in its most developed form, requires 10 or 11 spacetime dimensions for anomaly cancellation and quantum consistency. The extra dimensions are typically assumed to be compactified at a scale R , with Kaluza–Klein (KK) excitations at masses $m_n \sim n/R$.

Three broad classes of extra-dimensional models have been studied experimentally:

- (i) **ADD** (Arkani-Hamed–Dimopoulos–Dvali): δ large flat extra dimensions with the fundamental scale M_D at the TeV scale.
- (ii) **RS** (Randall–Sundrum): one warped extra dimension with an exponential hierarchy between the Planck and TeV scales.
- (iii) **UED** (Universal Extra Dimensions): all Standard Model fields propagate in the bulk.

Despite extensive searches, no evidence for any of these has been found. This paper examines the status of extra dimensions from both the experimental side and the LoNalogy framework, and argues that LoNalogy provides a 4D-complete alternative in which the structural roles of extra dimensions are played by internal modular/fiber geometry.

2 Experimental Status

2.1 LHC constraints

The LHC Run 2 dataset ($\sqrt{s} = 13$ TeV, ~ 140 fb $^{-1}$) provides the strongest collider constraints on extra-dimensional models.

Model	Channel	Constraint
ADD (graviton emission)	monojet (ATLAS/CMS)	$M_D \gtrsim 5.5\text{--}11.2$ TeV
ADD (virtual graviton)	dijet angular (CMS)	$M_{TT} > 9.02$ TeV
RS (KK graviton)	diphoton (CMS)	$m_1 \gtrsim 2.2\text{--}5.6$ TeV
RS (KK graviton)	diphoton (ATLAS)	$m_1 > 4.5$ TeV ($k/\bar{M}_P = 0.1$)
Radion	ATLAS/CMS	$m_{\text{radion}} \gtrsim 3.1\text{--}3.2$ TeV
Microscopic BH	high-multiplicity (CMS)	$M_{\text{BH}} \lesssim 10.1$ TeV excluded
UED ($1/R$)	various	$1/R \gtrsim 1.4\text{--}1.5$ TeV

Minimal UED is ruled out when LHC constraints are combined with the dark matter relic density requirement.

2.2 Short-distance gravity

Tabletop gravity experiments constrain large flat extra dimensions directly. For $\delta = 1$ flat extra dimension, the current bound is $R < 44$ μm . For $\delta = 2$, ATLAS monojet data give $R < 3.8$ μm .

2.3 Summary of experimental status

Proposition 2.1 (Experimental verdict).

$$\boxed{\text{Low-scale literal extra dimensions: strongly constrained toward exclusion.}} \quad (2.1)$$

$$\boxed{\text{Planck-scale compactified extra dimensions: logically possible but experimentally inaccessible.}} \quad (2.2)$$

3 LoNalogy Is a 4D Theory

3.1 The core action

Definition 3.1 (LoNalogy core action). The LoNalogy master action is defined on a 4-dimensional spacetime:

$$S_{\text{LoN}} = S_{\text{grav}} + S_x + S_{\text{vis}} + S_A + S_X^{\text{eff}}, \quad (3.1)$$

where

$$S_{\text{grav}} + S_x = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} G_{xx}(x) \partial_\mu x \partial^\mu x - V(x) \right], \quad (3.2)$$

and the visible, portal, and bridge sectors are likewise written as 4D actions. No fifth or higher coordinate $x^5, x^6, \dots$ appears.

Remark 3.2. The spacetime integral is $\int d^4x$, not $\int d^{10}x$ or $\int d^{11}x$. LoNalogy is a 4D effective theory from the outset.

3.2 Internal dimensions vs. spacetime dimensions

Several large integers appear in the LoNalogy framework. It is essential to distinguish these from spacetime dimensions.

Proposition 3.3 (Internal dimensions are not spacetime coordinates). *The integers appearing in LoNalogy have the following origins:*

- (i) $\nu_{\text{br}} = 12$: the real dimension of the broken tangent space $SU(5)/S(U(3) \times U(2)) \cong \text{Gr}_2(\mathbb{C}^5)$, computed as $2 \cdot 3 \cdot 2 = 12$. This is an internal coset dimension, not a spacetime coordinate count.
- (ii) $d_{\text{fb}} = 7$: the fiber/bridge dimension in the three-sector decomposition. This counts internal degrees of freedom, not spatial directions.
- (iii) $N = 5$ (rank): selected by the Yukawa closure $\Lambda^2 V \otimes \Lambda^2 V \otimes V \rightarrow \det V$ and anomaly cancellation. This is a bundle rank, not a spacetime dimension.

None of these is an additional spacetime coordinate x^μ .

Remark 3.4. The confusion between internal and spacetime dimensions is the most common misreading of LoNalogy by those trained in string theory, where internal dimensions *are* literally extra spacetime coordinates (compactified).

4 De-Literalisation of Extra Dimensions

Theorem 4.1 (De-literalisation). *Every structural role played by extra dimensions in string/KK theory has a 4D counterpart in LoNalogy:*

Role	String/KK language	LoNalogy language
Gauge unification	Compactification on G/H	Rank-5 bundle on $X(2)$
Family structure	Fixed points / intersections	Three cusps $\{0, 1, \infty\}$
Coupling hierarchy	Warp factor $e^{-kr_c\pi}$	Bridge reciprocity $M_C\ \sigma_{\text{br}}\ = \Lambda_*$
KK tower	$m_n = n/R$	Holonomy Fourier modes on $u \in [0, 1)$
Moduli stabilisation	Flux compactification	Strict kernel $\nu_{\text{br}}(1-k)^2(1+k) = 2$
Hidden sector	Bulk / brane separation	$\mathfrak{I}$ -sector (idea), Schur complement

In each case, the LoNalogy construction lives entirely in 4D with internal (fiber/modular) structure, without introducing additional spacetime coordinates.

Proof. Each row is verified by direct comparison with the LoNalogy framework:

Gauge unification. String theory compactifies on a manifold G/H whose isometry group gives the gauge group. LoNalogy obtains $SU(3) \times SU(2) \times U(1)$ from the rank-5 bundle on $X(2)$ via the anomaly-free decomposition $5 = 3 + 2$. No compact manifold with spatial coordinates is needed.

Family structure. In string theory, three generations arise from intersection numbers or fixed points of the compact manifold. In LoNalogy, three generations correspond to the three cusps $\{0, 1, \infty\}$ of $X(2)$. These are modular-geometric features, not spatial locations.

Coupling hierarchy. The RS model generates the Planck/TeV hierarchy via the warp factor $e^{-kr_c\pi}$. LoNalogy generates the hierarchy via the bridge reciprocity law $M_C\|\sigma_{\text{br}}\| = \Lambda_*$ and the strict kernel equation, which fixes the modular parameter algebraically.

KK tower. In KK theory, the tower $m_n = n/R$ arises from Fourier decomposition on the compact direction. In LoNalogy, the holonomy variable $u \in [0, 1)$ plays the analogous role (cf. the universal projected-determinant theorem), but it is a *fiber* coordinate, not a spacetime coordinate.

Moduli stabilisation. String theory requires flux compactification to fix the moduli (sizes and shapes of extra dimensions). LoNalogy fixes its moduli via the strict kernel equation $\nu_{\text{br}}(1-k)^2(1+k) = 2$, which has a unique solution k_{ker} by cubic analysis.

Hidden sector. In braneworld models, the hidden sector lives on a separate brane separated in the extra dimension. In LoNalogy, the hidden sector $\mathfrak{I}$ is separated by the Schur complement structure, not by spatial distance. $\square$

Corollary 4.2 (Extra dimensions de-literalised).

$$\boxed{\text{Extra dimensions are not denied; they are de-literalised.}} \quad (4.1)$$

The structural content of extra dimensions—unification, family structure, hierarchy, tower, moduli, hidden sector—is preserved in LoNalogy but housed in internal modular/fiber geometry rather than additional spacetime coordinates.

5 Consistency with AdS/CFT

Proposition 5.1 (Compatibility with holographic duality). *The de-literalisation of extra dimensions in LoNalogy is consistent with, and in fact anticipated by, the AdS/CFT correspondence.*

In the RS model, the 5D warped geometry AdS_5 is dual to a 4D strongly-coupled composite theory via the AdS/CFT dictionary. The KK graviton resonances in 5D correspond to composite spin-2 resonances in 4D. The PDG review explicitly notes this duality.

LoNalogy chooses the “4D side” of this duality from the outset. The bridge structure (positive frame operator $\Delta_{\text{br}} = \Sigma^\dagger \Sigma$, sech^2 weight, Schur complement reduction) plays the role of the 4D composite dynamics that is dual to the 5D bulk.

6 What the LHC Should Look For

Proposition 6.1 (LoNalogy predictions vs. KK tower). *If LoNalogy is correct, the LHC should not expect to see a KK tower ($m_n = n/R$). Instead, the signatures of new physics are 4D bridge/portal/hidden-sector phenomena:*

- (i) *Dark matter candidate χ at 338–348 GeV (bridge sector, spin-independent cross section predicted).*
- (ii) *Light portal scalar A' at $m_{A'} = 0.204 \text{ GeV}$ (portal amplitude $\epsilon_{\text{port}} = 12Q_{\text{ker}}^4$).*
- (iii) *Proton decay at the LoNalogy-predicted rate (τ_p from the GUT scale M_X).*

These are 4D predictions, not extra-dimensional signatures. The question “can the LHC see extra dimensions?” is, from the LoNalogy perspective, a category error.

The correct question: can the LHC see the 4D bridge/portal structure?

(6.1)

7 Final Verdict

Statement	Status
Literal extra spacetime dimensions exist	Unconfirmed
Low-scale ($\sim \text{TeV}$) extra dimensions	Strongly constrained
Planck-scale compactified extra dimensions	Logically possible, experimentally inaccessible
LoNalogy requires literal extra dimensions	No
LoNalogy reproduces the structural roles of extra dimensions	Yes (via internal geometry)

The shortest answer:

Extra dimensions: not denied, but de-literalised. LoNalogy is 4D-complete.

(7.1)

Conclusion

The experimental status is clear: no evidence for extra dimensions has been found, and the simplest low-scale models are increasingly excluded. The LoNalogy framework provides an explanation for this null result: literal extra spacetime coordinates were never needed.

The structural content of extra dimensions—gauge unification, family structure, coupling hierarchy, moduli stabilisation, hidden sectors—is entirely captured by the internal modular/fiber geometry of $X(2)$: the rank-5 bundle, three-cusp family, bridge reciprocity, strict kernel, and Schur complement. The core action is 4-dimensional.

This is not a denial of extra dimensions but their *de-literalisation*: the physics that string theory housed in extra spatial coordinates, LoNalogy houses in internal geometry. The two descriptions may ultimately be dual (cf. AdS/CFT), but LoNalogy demonstrates that the 4D description is self-contained and predictive.

In one sentence:

Extra dimensions are the shadow cast by internal modular geometry onto the language of spacetime coordinates.

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References

- [1] M. Nakamura, *LoNalogy v87.1*, Zenodo (2026), doi:10.5281/zenodo.19115338.
- [2] M. Nakamura, *Yang–Mills Existence and Mass Gap*, Zenodo (2026), doi:10.5281/zenodo.19183838.
- [3] R. L. Workman *et al.* (Particle Data Group), *Review of Particle Physics*, Prog. Theor. Exp. Phys. (2025).
- [4] N. Arkani-Hamed, S. Dimopoulos, and G. Dvali, *The Hierarchy Problem and New Dimensions at a Millimeter*, Phys. Lett. B **429**, 263 (1998).
- [5] L. Randall and R. Sundrum, *A Large Mass Hierarchy from a Small Extra Dimension*, Phys. Rev. Lett. **83**, 3370 (1999).